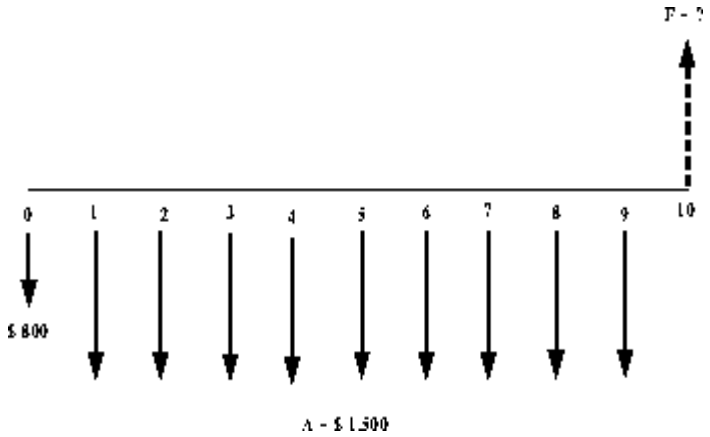
Set 1

Problem 1

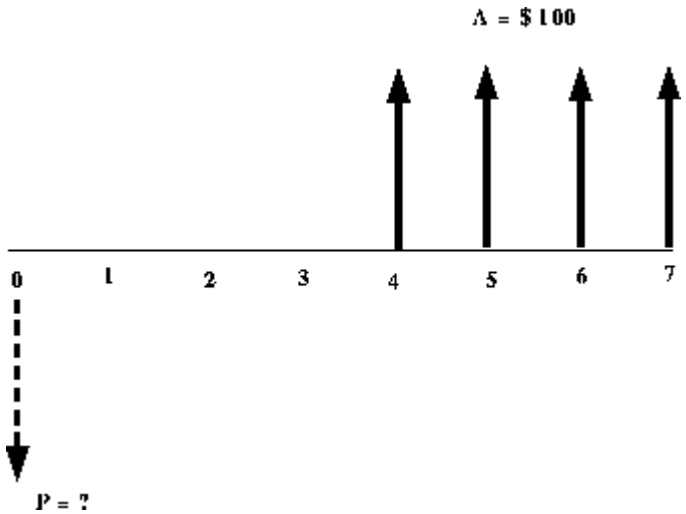
If you make the following series of deposits at an interest rate of 10% compounded annually, what would be the total balance at the end of 10 years?



A. $ F = 22,256
B. $ F = 24,481
C. $ F = 24,881
D. $ F = 25,981

Problem 2

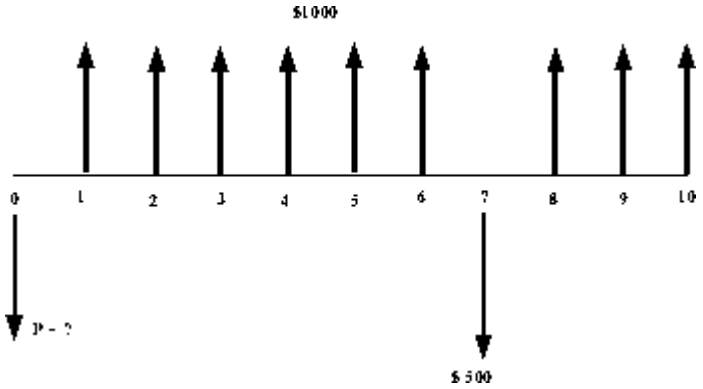
In computing the equivalent present worth of the following cash flow series at period 0, which of the following expression is incorrect?



A. P = $ 100(P/A,i,4)(P/F,i,4)
B. P = $ 100(F/A,i,4)(P/F,i,7)
C. P = $ 100(P/A,i,7) - $ 100(P/A,i,3)
D. P = $100[(P/F,i,4) + (P/F,i,5) + (P/F,i,6) + (P/F,i,7)]

Problem 3

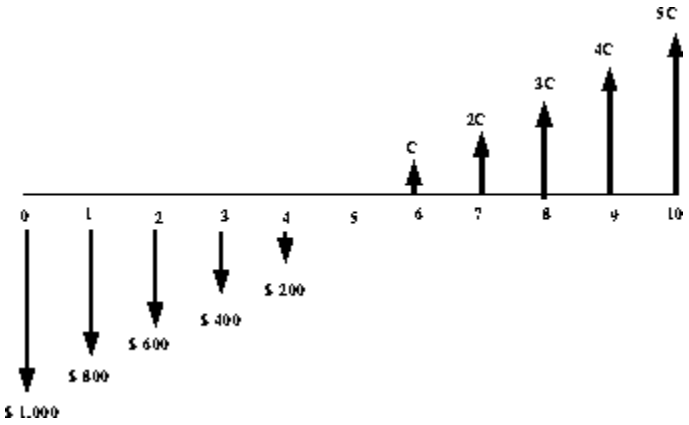
To withdraw the following $ 1,000 payment series in the figure, determine the minimum deposit (P) you should make now if your deposits earn an interest rate of 10 % compounded annually. Note that you are making another deposit at the end of year 7 in the amount of $ 500. With the minimum deposit P, your balance at the end of year 10 should be 0.



A. P = $ 4,912
B. P = $ 4,465
C. P = $ 5,374
D. P = $ 5,912

Problem 4

Consider the cash flow series shown below. What value of C makes the inflow series equivalent to the outflow series at an interest rate of 12% compounded annually?



A. C = $ 200
B. C = $ 282.7
C. C = $ 394.65
D. C = $ 458.90

Problem 5

You just received credit card applications from two different banks. The interest terms on your unpaid balance are as follows:

- Bank A: 15% compounded monthly
- Bank B: 14.8% compounded daily

Which of the following statements is incorrect?

A. The effective annual interest rate for bank A is 16.075%
B. The nominal interest rate for bank B is 14.8%
C. Bank A's term is a better deal because you will pay less interest on your unpaid balance.
D. The effective monthly interest rate for Bank A is 1.25%

Problem 6

John secured a home improvement loan in the amount of $ 10,000 from a local bank at an interest rate of 9% compounded monthly. He agreed to pay the loan in 60 equal monthly installments. Right after the 24th payment, John wishes to pay off the remainder of the loan in a lump sum amount. What is the payment size?

A. $ 7,473
B. $ 6,000
C. $ 6,528
D. $ 7,710

Problem 7

Vi Wilson is interested in buying an automobile priced at $ 18,000. She can come up with a down payment in the amount of $ 3,000 from her personal savings. The remainder of the loan will be financed over a period of 36 months from the dealer at an interest rate of 6.25% compounded monthly. Which of the following statement is correct?

A. The dealer's annual percentage rate (APR) is 6.432%.
B. The monthly payment can be calculated by A = $15,000(A/P,6.25%,3)/12
C. The monthly payment can be calculated by A = $15,000(A/P,6.24%/12,36)
D. The monthly payment can be calculated by A = $15,000(A/P,6.432%,3)/12

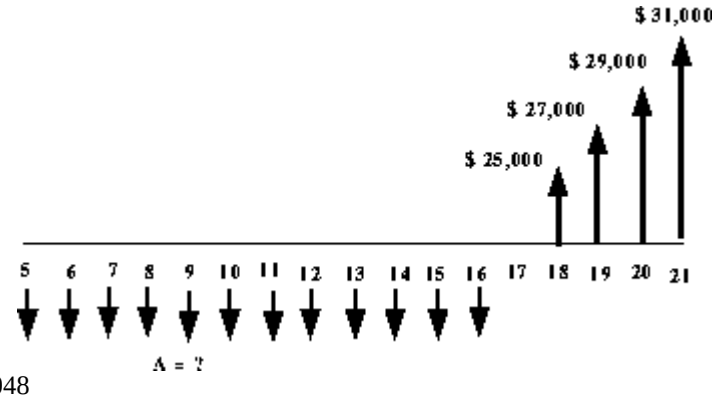
Problem 8

What is the future worth of an equal quarterly payment series of $2,500 for 10 years if interest rate is 9% compounded monthly?

A. F = $ 158,653
B. F = $ 151,930
C. F = $ 154,718
D. F = $ 160,058

Problem 9

A couple is planning to finance their 5-year-old daughter's college education. They were able establish a college fund that earns 8% compounded annually. What annual deposit must be made from the daughter`s 5th birthday to her 16th birthday to meet the future college expenses shown in the following figure. Assume that today is her 5th birthday.



A. A = $ 3,048
B. A = $ 5,893
C. A = $ 4,494
D. A = $ 4,854

Problem 10

You are considering two savings plans for your future retirement.

- **Option 1:** Deposit $1,000 at the end of each quarter for the first 10 years. At the end of 10 years. At the end of 10 years, make no further deposits, but leave the amount accumulated at the end of 10 years for the next 15 years.
- **Option 2:** Do nothing for the first 10 years. Then deposit $6,000 at the end of each year for the next 15 years.

If your deposits or investments earn at an interest rate of 6% compounded quarterly, which of the following statement is correct? With Option 2, when compared with Option 1, you will be able to accumulate

E. $ 7,067 more at the end of 25 years from now
F. $ 8,523 more at the end of 25 years from now
G. $ 14,757 less at the end of 25 years from now
H. $ 13,302 less at the end of 25 years from now

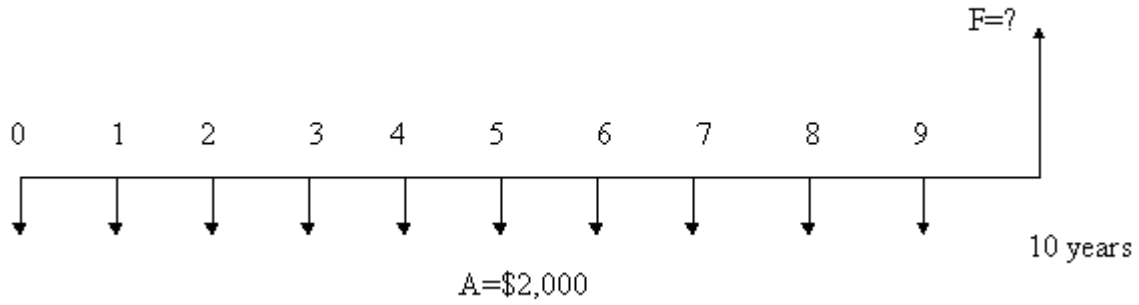
Solutions

1. B
2. A
3. C
4. D
5. C
6. C
7. C
8. D
9. C
10. B

Set 2

Problem 1

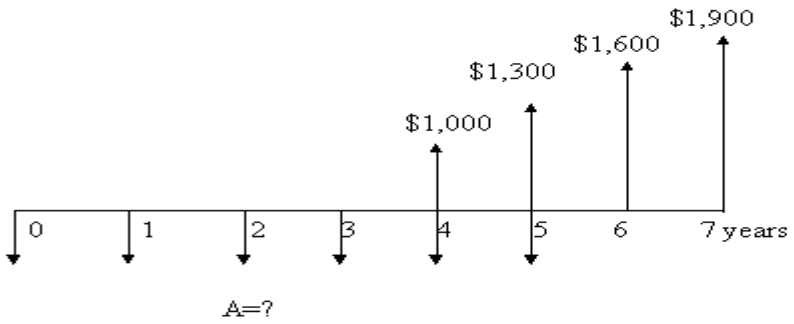
What is the future worth (at the end of year 10) of a cash flow series of 10 equal annual deposits of $2,000 if all deposits are made at the beginning of each year starting today (year 0) at 9% annual interest? Choose the correct range.



A. less than $33,118
B. between $33,119 and $33,122
C. between $33,123 and $33,124
D. between $33,125 and $33,126
E. more than $33,127

Problem 2

Consider the cash flow series shown below. Find out the required equal annual deposits (end of year) in a bank to generate the cash flows from year 4 through year 7.
Assume that an interest rate is 10% compounded annually. The value of A should be:



A. less than $693
B. between $694 and $696
C. between $697 and $699
D. between $700 and $702
E. more than $702

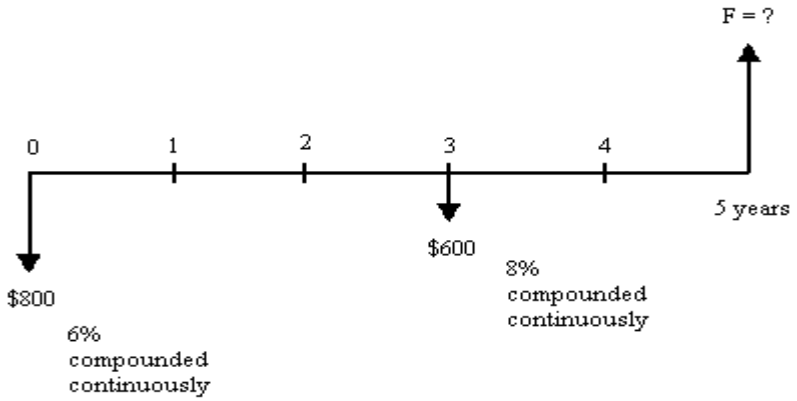
Problem 3

You bought a car by securing a loan in the amount of $20,000 from Auburn Bank at an interest rate of 9% compounded monthly. You agreed to pay off the loan in 48 equal monthly installments (each payment occurs at the end of each month). Immediately after 36th payment, you want to pay off the remainder of the loan in a lump sum amount, what should this amount be?

A. less than $5,693
B. between $5,693 and $5,695
C. between $5,696 and $5,698
D. between $5,699 and $5,701
E. more than $5,703

Problem 4

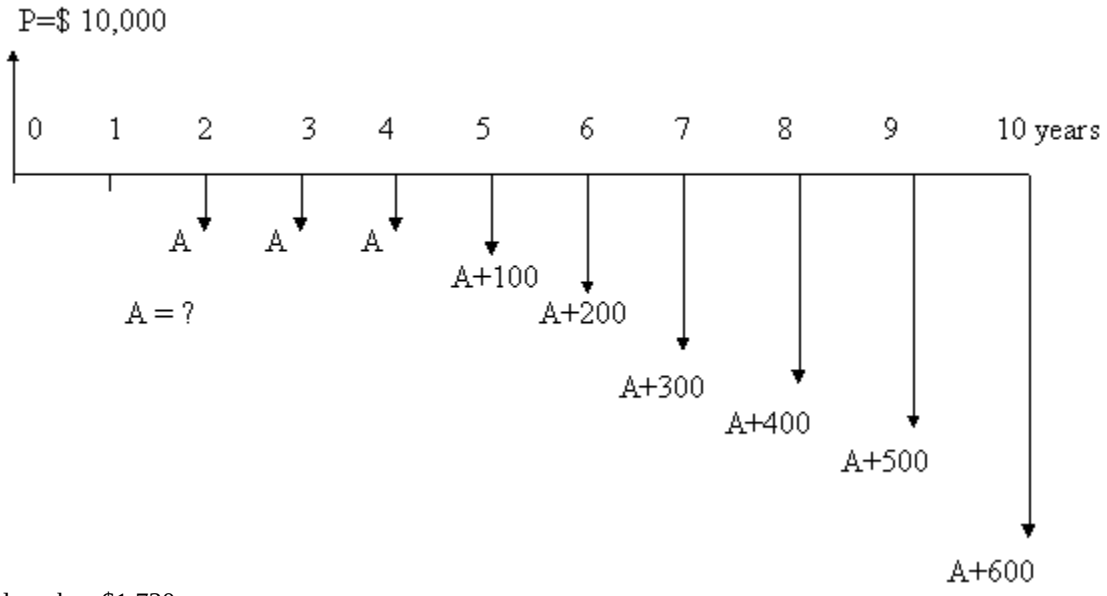
Consider the cash flow series shown below. What is the total future value of cash outflows at the end of 5 years with the changing interests specified?



A. less than $1,813
B. between $1,814 and $1,817
C. between $1,818 and $1,822
D. between $1,823 and $1,826
E. more than $1,827

Problem 5

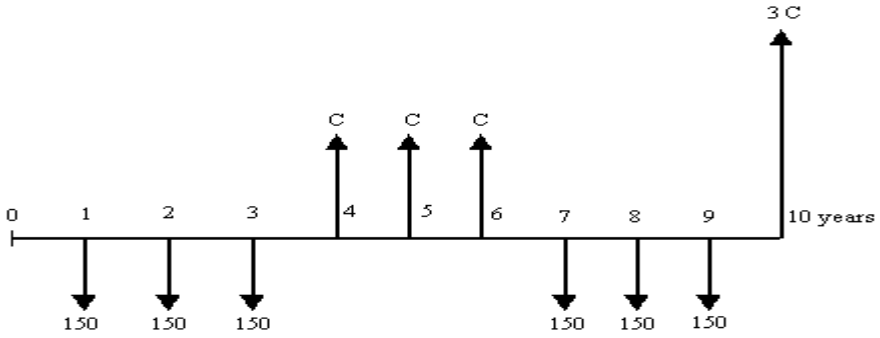
If you borrow $10,000 at 10% compounded annually with the repayment schedule below, what is the amount A?



A. less than $1,730
B. between $1,731 and $1,735
C. between $1,736 and $1,740
D. between $1,741 and $1,745
E. more than $1,746

Problem 6

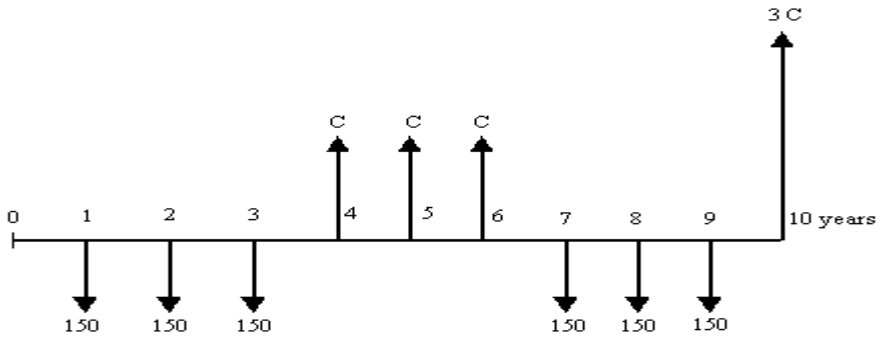
Consider the cash flow series shown below. What value of C makes the inflow series equivalent to the outflow series at an interest rate of 6% compounded annually?



A. less than $160
B. between $161 and $170
C. between $171 and $180
D. between $181 and $190
E. more than $191

Problem 7

Consider the following cash flows:



What is P equal to if i=10% compounded annually?

A. 2,000(P/A,10%,8)(P/F,10%,1) + 1,000(P/G,10%,6)(P/F,10%,3)
B. 2,000(P/A,10%,8)(P/F,10%,1) + 1,000(P/G,10%,5)(P/F,10%,4)
C. 2,000(P/A,10%,8)(P/F,10%,1) + 1,000(P/G,10%,4)(P/F,10%,4) + 4,000(P/F,10%,9)
D. 2,000(P/A,10%,8)(P/F,10%,1) + 1,000(P/G,10%,5)(P/F,10%,3) + 4,000(P/F,10%,9)

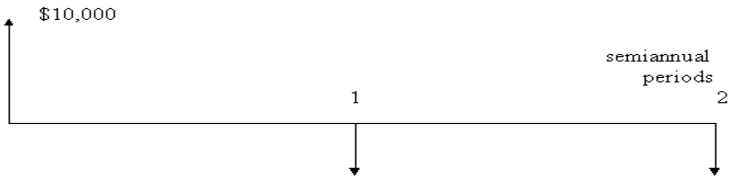
Problem 8

You have $5,000 to invest in a financial security. From a financial point of view, which of the following is the worst deal?

A. 12% compounded annually.
B. 11.8% compounded semi-annually.
C. 11.5% compounded quarterly.
D. 11.2% compounded daily.

Problem 9

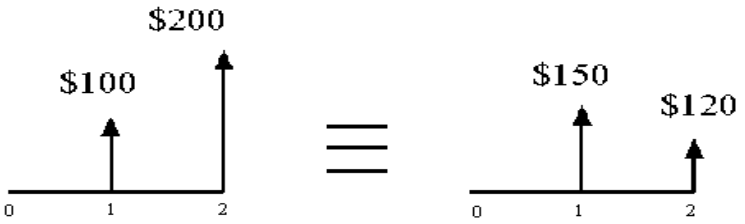
You want to borrow $10,000 from a local bank, which is to be repaid in 2 equal semiannual installments. The loan officer initially offered an interest rate of 12% compounded monthly. However, you were able to negotiate that interest be compounded semiannually instead of monthly. With this negotiation, how much do you save in total interest payments over the loan life?



A. less than $20
B. between $21 and $25
C. between $26 and $30
D. more than $31

Problem 10

What interest rate would make the following two cash flows equivalent?



A. 0% < iø 15%
B. 15% < iø 30%
C. 30% < iø 45%
D. 45% < iø 60%
E. 60% < iø 100%

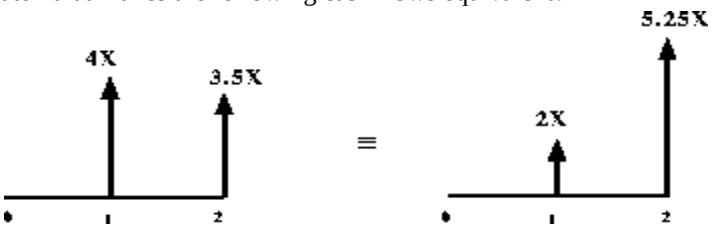
Solutions:

1. B
2. R
3. A
4. D
5. A
6. C
7. D
8. D
9. B
10. D

Set 3

Problem 1

Find the interest rate *i* that makes the following cash flows equivalent?



A. *i* = 0.125%
B. *i* = -0.125%
C. *i* = 12.5%
D. *i* = -12.5%

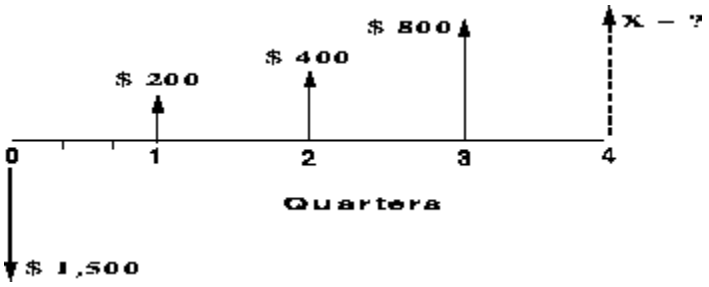
Problem 2

You are considering investing $1,000 in the stocks of two companies. Company A's stock is expected to grow at an annual average rate of 11% for the first 5 years and 15% for the next 5 years. Company B's stock is expected to grow at an annual average rate of 14% for the first 4 years and 12% for the next 6 years. If you plan to keep both stocks for the next 10 years, which of the following statements is correct?

A. Both stocks have the same future worth at the end of year 10
B. Company A's stock has a $55 higher future worth at the end of year 10
C. Company B's stock has a $40 higher future worth at the end year 10
D. Company B's stock has a $26 higher future worth at the end year 10

Problem 3

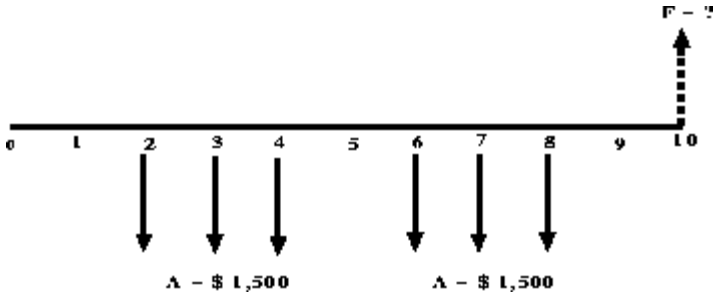
You just deposited $1,500 into a savings account that pays 9% interest, compounded monthly. If you intend to take out $200 at the end of the first quarter, $400 at the end of the second quarter, and $800 at the end of the third quarter, what is the maximum amount that you can withdraw at the end of the 4th quarter?



A. X = $176
B. X = $184
C. X = $191
D. X = $200

Problem 4

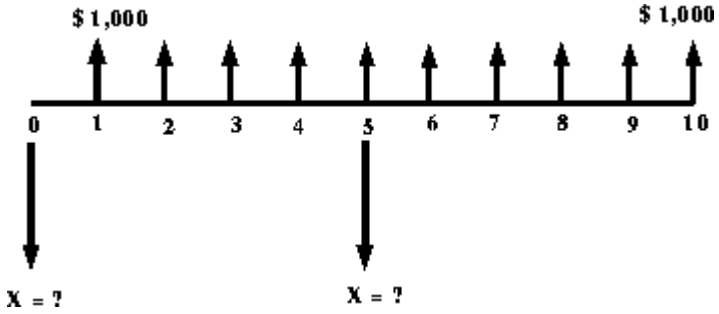
Find the future worth F of the following cash flow at 10% interest, compounded annually.



A. F = $14,008
B. F = $13,237
C. F = $17,219
D. F = $14,803

Problem 5

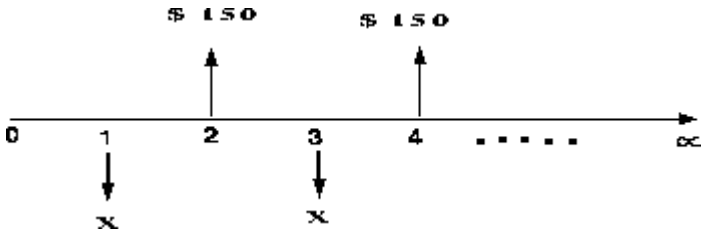
Determine the two equal deposits (the first deposit required now and the second deposit at the end of year 5) so that you can withdraw $1,000 at the end of each year for the next 10 years. Assume that money can earn 10% interest, compounded annually



A. X = $3,791
B. X = $3,072
C. X = $3,605
D. X = $6,145

Problem 6

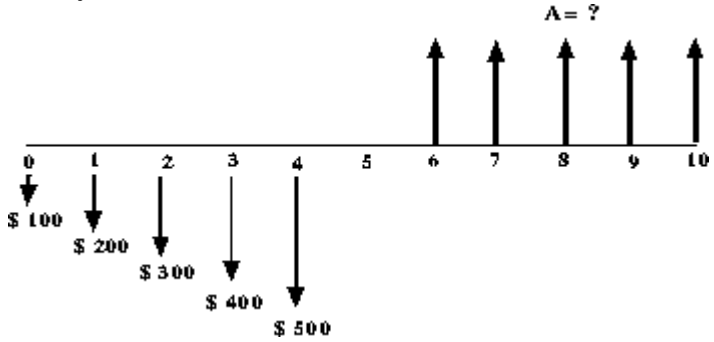
If you want to withdraw $150 at the end of every even year (i.e., years 2, 4,..), how much should you deposit at the end of every odd year (i.e., years 1, 3, ...)? Assume that the interest rate is 20%, compounded annually.



A. X = $150
B. X = $137.5
C. X = $125
D. X = $100

Problem 7

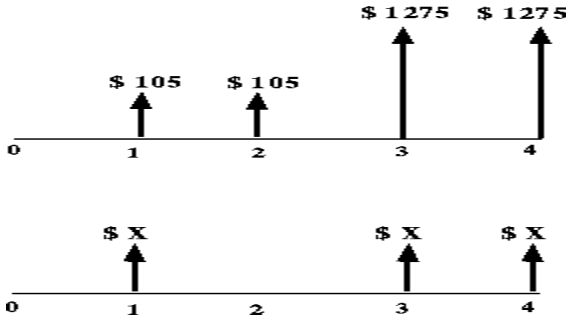
Find the equal payment amount A, that makes the inflow series equivalent to the outflow series at i = 12%, compounded annually.



A. A = $489
B. A = $547
C. A = $600
D. A = $636

Problem 8

Find the value X, that makes the following two cash flow series equivalent at i = 10%?



A. X = $636
B. X = $765
C. X = $858
D. X = $920

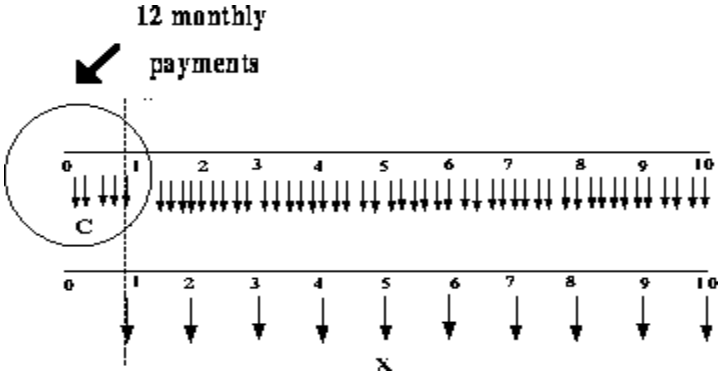
Problem 9

If you borrowed $12,000 payable in 6 equal annual installments, what would be the principal payment due at the end of year 1. Use i = 13% compounded annually.

A. $2,000
B. $1,560
C. $1,370
D. $1,440

Problem 10

Suppose you were to receive C at the end of each month for the next 10 years (Option 1). Alternatively, you can receive an equal end-of-year amount X, over the next 10 years (Option 2). What value of X would you prefer option 2 over option 1 at an interest rate of 12%, compounded monthly?



A. X greater than 12C(F/A, 12.68%, 10)(A/F, 12.68%, 10)
B. X greater than C(F/A, 1%, 12)
C. X greater than C(F/A, 1%, 120)(A/F, 12%,10)
D. X greater than C(F/A, 1%, 120)(A/F, 12.75%,10)

Solutions:

1. D
2. B
3. C
4. D
5. A
6. C
7. B
8. C
9. D
10. B

Set 1

Problem 1

Given the following financial data, determine the Conventional Payback Period

- Investment cost at n=0: $10,000.
- Investment cost at n=1: $15,000
- Useful life: 10 Years
- Salvage value (at the end of 11 years): $5,000
- Annual revenues: $12,000 per year
- Annual expenses: $4,000 per year
- MARR = 10%

(Note: The first revenues and the expenses will occur at the end of year 2.)
In the following, the ranges indicate the actual time interval where the Payback will occur from the inception of the project.

A. Payback Period: greater than 3 and less or equal 4
B. Payback Period: greater than 4 and less or equal 5
C. Payback Period: greater than 5 and less or equal 6
D. Payback Period: greater than 6 and less or equal 7

Problem 2

The City of Auburn has decided to build a softball complex and the city council has already voted to fund the project at the level of $800,000 (initial capital investment). The city engineer has collected the following financial information for the complex project.

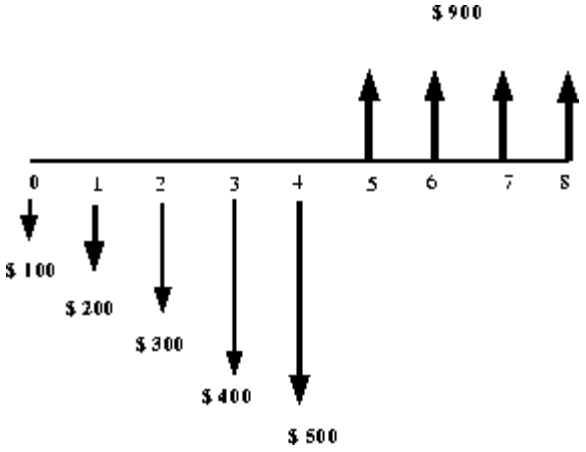
- Annual upkeep costs: $120,000
- Annual utility costs: $13,000
- Renovation costs: $50,000 for every 5 years
- Annual team user fees (revenues): $32,000
- Useful life: Infinite
- Interest rate: 8%

If the city can expect 40,000 visitors to the complex each year, what should be the minimum ticket price per person so that the city can breakeven?
In answering the question, identify the range that contains the solution.

A. Price: greater than $2.50 and less or equal $3.00
B. Price: greater than $3.00 and less or equal $3.50
C. Price: greater than $3.50 and less or equal $4.00
D. Price: greater than $4.00 and less or equal $4.50

Problem 3

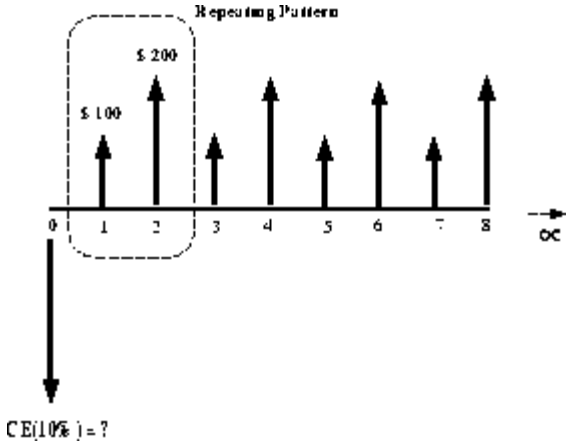
Find the present worth of the following cash flow series at an interest rate of 9%.



A. PW(9%): greater than $640 and less or equal $700
B. PW(9%): greater than $770 and less or equal $800
C. PW(9%): greater than $860 and less or equal $890
D. PW(9%): greater than $900 and less or equal $930

Problem 4

Find the capitalized equivalent worth of the following project cash flow series at an interest rate of 10%.



A. CE(10%) = $ 1,476
B. CE(10%) = $ 1,500
C. CE(10%) = $ 3,000
D. CE(10%) = $ 1,753

Problem 5

Consider the following two investment situations:

In 1970, when Wal-Mart Stores, Inc. went public, an investment of 100 shares cost $1,650. That investment would have been worth $2,991,080 after 25 years. The Wat-Mart investors' rate of return would be around 35%
In 1980, if you bought 100 shares of Fidelity Mutual Funds, it would cost $5,245. That investment would have been

worth $80,810 after 15 years.

Which of the following statements is correct?

A. If you bought only 50 shares of Wal-Mart stocks in 1970 and kept it for 25 years, your rate of return would be 0.5 x 35%.
B. The investors in Fidelity Mutual Funds would have made a profit at an annual rate of 30%.
C. If you bought 100 shares of Wal-Mart shares in 1970 but sold them after 10 years. Then immediately, put all your proceeds in buying the Fidelity Funds. After 15 years, the total worth of your investment would be around $511,140
D. None of the above

Problem 6

You purchased a CNC machine for $34,000. It is expected to have an useful life of 10 years and a salvage value of $3,000. At i=15%, what is the annual capital cost of this machine?

A. $6,775
B. $3,100
C. $6,627
D. $3,400

Problem 7

You are considering an investment that costs $2,000. It is expected to have a useful life of 3 years. You are very confident about the revenues during the first two years but you are unsure about the revenue in year 3. if you hope to make at least 10% rate of return on your investment ($2,000), what should be the minimum revenue in year 3.

Year	Cash Flow
0	$-2,000
1	$1,000
2	$1,200
3	$ X

A. X=$220
B. X=$132
C. X=$300
D. X=$274

Problem 8

You need a lathe for your machine shop for 10 years. You narrowed down to two models: Kendall and Toyota. You also collected the following financial data:

	Kendall	Toyota
First Cost	$25,000	$32,000
O & M Cost	$11,000 / yr	$9,700 / yr
Useful life	10 years	14 years
Salvage	$3,000	$2,000

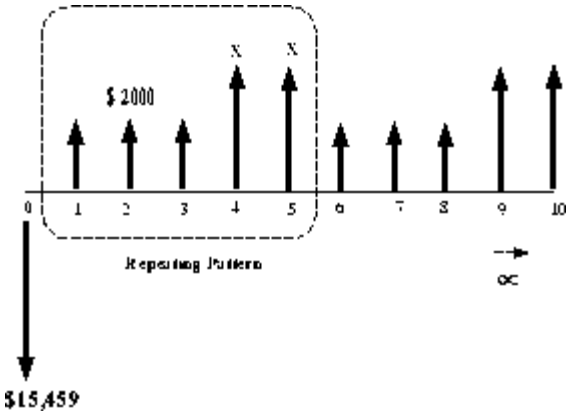
(Note that the salvage values represent the values at the end of useful life.)

If your interest rate is 12%, what should be the salvage value of the Toyota model at the end of 10 years so that you would be indifferent between the two models?

A. $1,240
B. $1,540
C. $1,610
D. $1,927

Problem 9

The following infinite cash flow series has a rate of return of 15%. Determine the unknown value X.



A. A = $2,500
B. A = $3,000
C. A = $3,200
D. A = $1,967

Problem 10

The following table contains the summary of how a project's balance changes over its 5-year service life at 10% interest (MARR).

End of period	Project Balance
0	$-1,000
1	$-1,500
2	$600
3	$900
4	$1,500
5	$2,000

Which of the following statements is incorrect?

A. The required additional investment at the end of period 1 is $500.
B. The net present worth of the project at 10% interest is $1,242.
C. The net future worth of the project at 10% interest is $2,000
D. The rate of return of the project should be greater than 10%.

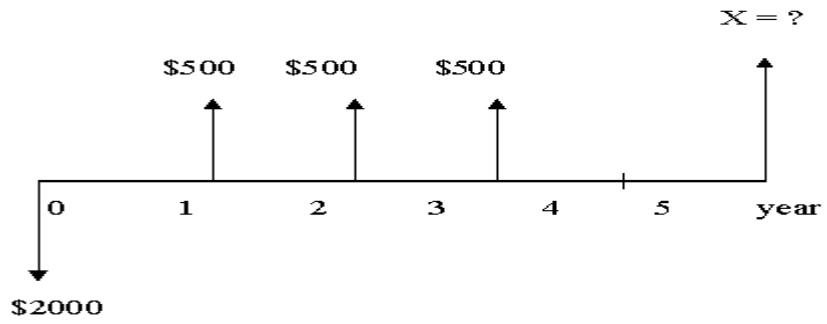
Solutions:

1. B
2. D
3. C
4. A
5. C
6. C
7. B
8. D
9. B
10. A

Set 2

Problem 1

A company invests $2000 in a project over five years. At the end of every year, for the first three years the project generates $500. At the end of the fourth year the project generates no money. At the end of the fifth year, the project is terminated. How much must the project generate at the end of the fifth year to realize a 13% return on the initial investment?



A. X <= $1400
B. $1400 < X <= $1430
C. $1430 < X <= $1490
D. $1490 < X <= $1520
E. X > $1520

Problem 2

Which of the following investment options will maximize your future wealth at the end of 20 years? Assume any funds that remain invested will earn a nominal rate of 12% compounded monthly

A. deposit $5000 now
B. deposit $80 at the end of each month for the first 10 years
C. deposit $50 at the end of each month for 20 years
D. deposit a lump sum in the amount of $15,000 at the end of year 10

Problem 3

Geo-Star Manufacturing Company is considering a new investment in a punch press machine that will cost $100,000 and has an annual maintenance cost of $10,000. There is also an additional overhauling cost of $20,000 for the equipment once every four years. Assuming that this equipment will last infinitely under these conditions, what is the capitalized equivalent cost of this investment at an interest rate of 10%?

A. CE (10%) <= $236100
B. $236100 < CE (10%) <= $238100
C. $238100 < CE (10%) <= $240100
D. $240100 < CE (10%) <= $242100
E. CE (10%) > $244100

Problem 4

Company X has been contracting its overhauling work to Company Y for $40,000 per machine per year. Company X estimates that by building a $500,000 maintenance facility with a life of 15 years and a salvage value of $100,000 at the end of its life, they could handle their own overhauling at a cost of only $30,000 per machine per year. What is the minimum annual number of machines that Company X must operate to make it economically feasible to build its own facility? (Assume an interest rate of 10%)

A. 1 <= N <= 5
B. 6 <= N <= 7
C. 8 <= N <= 9
D. 10 <= N <= 11
E. 11 <= N <= 13

Problem 5

Right now you have $10,000 to invest over 5 years. The interest rate in the United States is 10%. The interest rate in France is 12% for invested Francs. The interest rate in Japan is 9% for invested Yen. Assume that these interest rates are expected to remain unchanged over the next 5 years. The current as well as the expected exchange rates is given below.

Current Exchange Rates	Expected exchange rates 5 years from now
1$ = 6 Francs	1$ = 8 Francs
1$ = 125 Yen	1$ = 100 Yen

Which of the following options (if any) will maximize your wealth in US$ at the end of 5 years?

A. Investing in the United States
B. Investing in France
C. Investing in Japan
D. Not enough information to compare

Problem 6

Company X is considering the purchase of a helicopter for connecting services between their base airport and the new inter-county airport being built about 30 miles away. It is believed that the chopper will be needed only for 6 years until the Rapid Transit Connection is phased in. The estimates on two types of helicopters under consideration, The Whirl 2 B and The ROT 8, are given below:

	The Whirl 2 B	The ROT 8
First Cost	$95,000	$120,000
Annual Maintenance	$3,000	$9,000
Salvage Value	$12,000	$25,000
Useful life in years	3	6

Assuming that the Whirl 2B will be available in the future with identical costs, what is the annual cost advantage of selecting The Whirl 2B ? (Use an interest rate of 10 %)

A. cost more than $4,000
B. cost between $4,000 and $3,000
C. save between $3,000 and $2000
D. save more than $4,000
E. none of the above

Problem 7

Your company needs a machine for the next 7 years and you have two choices (assume an annual interest rate of 15%):

Machine A costs $100,000 and has an annual operating cost of $47,000. Machine A has a useful life of 7 years and a salvage value of $15,000.

Machine B costs $150,000 and has an annual operating cost of $30,000. Machine B has a useful life of 5 years and no salvage value. However the life of Machine B can be extended by two years with a certain amount of investment. If Machine B's life is extended it will still cost $30,000 annually to operate and still have no salvage value.

Which of the following values is the most you would pay at the end of year 5 to extend the life of Machine B by two years?

A. $100,000
B. $50,000
C. $40,000
D. $30,000
E. $15,000

Problem 8

The following information on 4 mutually exclusive projects is given below:

Project	Investment at year 0	IRR
A	$1,000	56%
B	$1,200	67%
C	$1,500	43%
D	$2,500	49%

All four projects have the same service life, and require investment in year 0 only. Suppose that you are provided with the following additional information about incremental rates of return between projects.

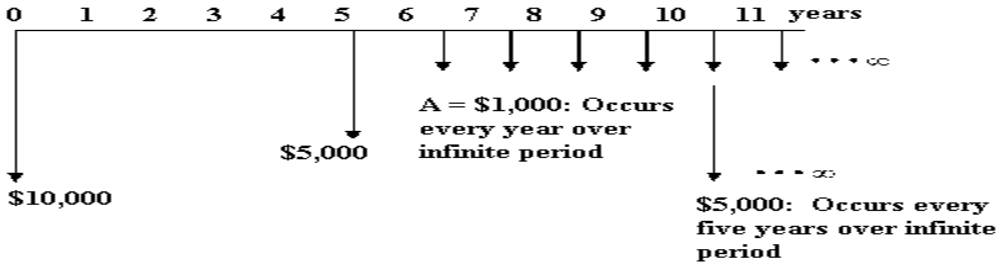
IRR (B - A) = 85%, IRR (D - C) = 25%, IRR (B - C) = 30%, and IRR (A - D) = 50 %

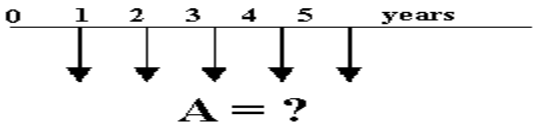
Which project would you choose based on the rate return criterion at a MARR of 29%?

A. Project A
B. Project B
C. Project C
D. Project D
E. Impossible to determine with the given information

Problem 9

Compute the annual equivalent cost of the cash flow series given below. Interest rate is 10%.





A. $2,100 < A <= $2,200
B. $2,200 < A <= $2,300
C. $2,300 < A <= $2,400
D. $2,400 < A <= $2,500
E. $2,500 < A <= $2,600

Problem 10

You are considering two types of electric motors to power an assembly line in your factory. The financial information and the operating characteristics of the two motors are given below. If you operate the assembly line for 6,000 hours annually, what is the total cost savings per operating hour associated with the more efficient brand (Brand B) at an interest rate of 10%? The motor is required for a period of 5 years.

	Brand A	**Brand B**
Price	$9,000	$12,000
Salvage after 5 years	$1,000	$1,500
Capacity	30 HP	30 HP
Efficiency	80%	85%

Current electricity price : $ 0.08 / kWh
Note: 1 HP = 0.7457 kW

A. less than or equal to 1 cent
B. greater than 1 cent but less than or equal to 3 cents
C. greater than 3 cents but less than or equal to 5 cents
D. greater than 5 cents but less than or equal to 7 cents
E. greater than 7 cents but less than or equal to 9 cents

Solutions:

1. D
2. B
3. D
4. B
5. C
6. A
7. D
8. C
9. D
10. B

Set 3

Problem 1

What is the present worth of the following cash flow series?

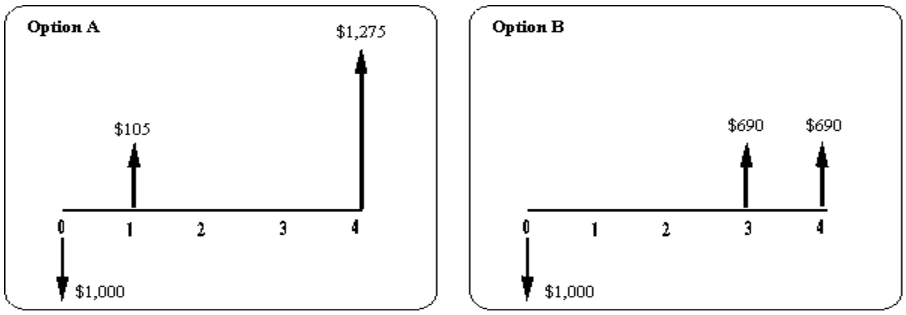
End of Year	**Cash Flow**
1	$1,900
2	$-1,000
3	$700

The interest rate for the first two years will be 12% compounded monthly and then it will change to 10% compounded quarterly.

A. $1,380
B. $1,400
C. $1,425
D. $1,600

Problem 2

Which of the following investment options would you choose at 10% interest, compounded annually?



A. Option A
B. Option A
C. Either (Both yield the same return)
D. Neither

Problem 3

General Mills Company (GMC) purchased a milling machine for $100,000 which it intends to use for the next 5 years. This machine is expected to save GMC $35,000 during the first operating year. Then, the annual savings are expected to decrease by *3% each subsequent year over the previous year* due to increased maintenance costs. Assuming that GMC would operate the machine for an average of 3,000 hours per year and that it would have no appreciable salvage value at the end of the 5-year period, determine the equivalent dollar savings per operating hour at 14% interest compounded annually.

A. $0.568 / hr
B. $0.827 / hr
C. $0.941 / hr
D. $1.369 / hr

Problem 4

Find the value P, such that the terminal project balance of the following cash flow is 0 at 10% interest compounded annually.

End of Year	Cash Flow	Project Balance
0	-P	-P
1	200	?
2	.5P	?
3	.25P	0

A. $475
B. $501
C. $455
D. $417

Problem 5

Find the value X such that the project's rate of return would be equal to 15%.

End of Year	Cash Flow
0	$-3,000
1	$1,330
2	X
3	$1,930

A. $840
B. $760
C. $620
D. $530

Problem 6

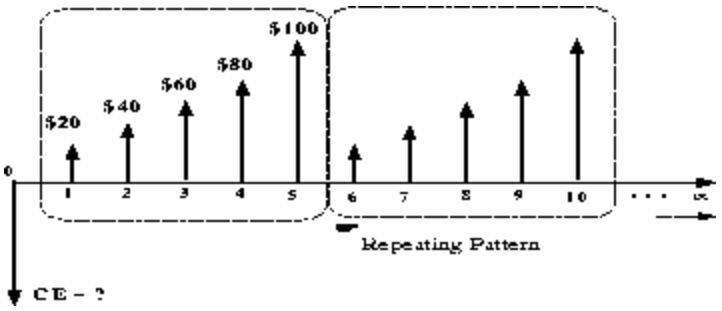
K.T. Labs bought a Gene gun for $25,000. The accounting department has estimated that the machine would have annualized capital cost of $3,880 over its 10-year service life. What salvage value was assumed in calculating the capital cost? The firm's interest rate is known to be 10%.
A. $3,201
B. $2,494
C. $2,750
D. $3,000

Problem 7

You are considering two investment options. In option A, you have to invest $5,000 now and $1,000 three years from now. In option B, you have to invest $3,500 now, $1,500 a year from now, and $1,000 three years from now. In both options, you will receive four annual payments of $2,000 each. (You will get the first payment a year from now.) Which of these options would you choose based on (i) the conventional payback criterion, and (ii) the present worth criterion, assuming 10% interest?
A. (i) Option A (ii) Option A
B. (i) Option B (ii) Option B
C. (i) Either (ii) Option A
D. (i) Either (ii) Option B

Problem 8

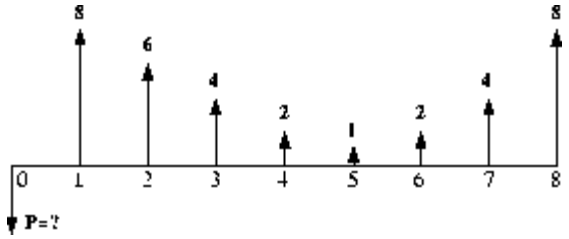
Find the capitalized equivalent worth of the following cash flow series at 10% interest.



A. $511
B. $445
C. $562
D. $482

Problem 9

Find the present worth P, of the following cash flow series at 10% interest.



A. $25.80
B. $28.75
C. $20.85
D. $24.10

Problem 10

Two 150 HP motors are being considered for installation at a water treatment plant. Both motors have a service life of 10 years at the end of which they have a negligible salvage value.

	Motor A	**Motor B**
Investment	$4,500	$3,600
Annual O & M cost (excluding power)	$675	$540
Full load efficiency	83%	80%

A. $900 < T <= $1,000
B. $1,000 < T <= $1,100
C. $1,100 < T <= $1,200
D. $1,200 < T <= $1,300

Solutions:

1. B
2. D
3. D
4. C
5. B
6. D
7. D
8. C
9. D
10. B

Set 1

Problem 1

Consider the following financial data for an investment project:

- Required capital investment at n=0: $100,000
- Project service life: 10 years
- Salvage value (at the end of 10 years): $10,000
- Annual revenues: $150,000 per year
- Annual O & M expenses: $50,000 per year
- Depreciation method for tax purposes: 7 - year MACRS
- Income tax rate: 40%

Determine the project cash flow at the end of year 10.

A. $60,000
B. $70,000
C. $66,000
D. $64,000

Problem 2

A local delivery company has purchased a delivery truck for $15,000. The truck will be depreciated under MACRS as 5-year property. The truck's market value (salvage value) is expected to decrease by $2,500 per year. It is expected that the purchase of the truck will increase its revenue by $10,000 annually. The O & M costs are expected to be $3,000 per year. The firm is in the 40% tax bracket and its MARR is 15%. If you plan to keep the truck for only two years, what would be the equivalent annual worth?

A. $473
B. $825
C. $638
D. $986

Problem 3

You are considering purchasing a CNC machine which costs $150,000. This machine will have an estimated service life of 10 years with a net after-tax salvage value of $15,000. Its annual after-tax O & M costs (considering depreciation tax shields) are estimated to be $50,000. To expect an 18% rate of return on investment (after-tax), what would be the required minimum after-tax revenues?

A. $94,568
B. $63,500
C. $92,435
D. $82,740

Problem 4

You purchased a computer which cost $50,000 four years ago. At that time, the computer was estimated to have a service life of 5 years with salvage value of $5,000. These estimates are still good. The property has been depreciated according to a 5-year MACRS property class. Now (at the end of year 4 from purchase) you are considering selling the computer at $8,000. What would be the book value that should be used in determining the taxable gains?

A. $8,640
B. $9,677
C. $11,520
D. $10,398

Problem 5

Consider the following two mutually exclusive service projects with individual lives of 2 years and 3 years respectively. (Mutually exclusive service projects will have identical revenues for each year of service) The interest rate is known to be 6%.

End of year	Project A	Project B
0	$-1000	$-1000
1	$-140	$-400
2	$-140 + 0 (salvage)	$-400
3	----	$-400 + $200 (salvage)

If the required service period is indefinite (infinite) and both projects can be repeated with the given costs above, which is the better choice?

A. Project A
B. Project B
C. Indifferent (either Project A or Project B)
D. Neither

Solutions:

1. C
2. C
3. D
4. C
5. A

Set 2

Problem 1

Consider the four mutually exclusive projects given below.

Project	Investment at year 0	IRR
A	$2,000	45%
B	$2,500	40%
C	$3,000	35%
D	$3,500	30%

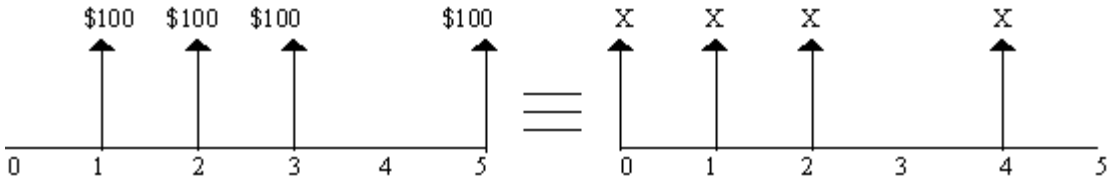
The service life for all the four projects is the same and investment is required only in year 0. Also, IRR(B-A) = 17%, IRR(A-D) = 15%, IRR(D-C) = 10%, IRR(B-C) = 20%

Which project would you prefer based on the rate of return criterion at a MARR of 13%?

A. A
B. B
C. C
D. D
E. Information insufficient to decide

Problem 2

The following two cash flow transactions are equivalent at an interest rate of 9%.



Find X

A. X < $90
B. $90 <= X < $94
C. $94 <= X < $98
D. $98 <= X < $102
E. None of the above

Problem 3

Farmer's National Bank gave you a loan of $30,000, to buy a car, at an interest rate of 12% compounded monthly. The agreement was to pay off the entire amount in 48 equal monthly installments (each payment occurs at the end of each month). After making the first 30 payments on time you missed the payments due at the end of the 31st and the 32nd months. Now at the end of the 33rd month you want to pay off the remaining amount of the loan in full. Assuming no penalty of charges for the missing payments, how much should you pay?

A. between $0 and $12,500
B. between $12,501 and $13,000
C. between $13,001 and $13,500
D. between $13,501 and $14,000
E. greater than $14,001

Problem 4

Five years ago you purchased 100 shares of Gillette stock at $30 per share. Now the current stock price is $85 per share. Assuming that the price of the stock is expected to continue an increasing trend in the future, which of the following statements is incorrect?

A. If you sell the stock right now, your annual rate of return on your Gillette investment will be less than 25%.
B. In order to realize a 25% annual rate of return, the current stock price should be about $92.
C. If you bought 200 shares 5 years ago instead of 100 shares, your rate of return would be doubled.
D. If you buy the Gillette stock at the current price of $85, and you expect the stock to appreciate at an annual rate of 12% (rate of return) in the future, it would take 6 years to double the price.

Problem 5

How long will it take for an investment to triple at an interest rate of 7%, compounded monthly?

A. between 0 and 10 years
B. between 10 and 12 years
C. between 12 and 14 years
D. between 15 and 16 years
E. between 17 and 50 years

Problem 6

Company J buys an equipment for $200,000. This equipment has a useful life of 10 years and a salvage value of $40,000. The company uses the straight line method to calculate the depreciation of the equipment. At the end of year 4, the company sells the equipment for $120,000. The tax rate is 40%

What is the net proceeds (after tax) from the sale of the equipment that the company reports?
(Note: *Assume that the half-year convention is not applied in depreciating the asset.*)
A. less than $120,000
B. between $120,001 and $122,000
C. between $122,001 and $124,000
D. between $124,001 and $126,000
E. greater than $126,001

Problem 7

ABC Corporation placed an asset in service 3 years ago. The company uses the *MACRS* method (7 year-life) for tax purposes and the *SOYD* method (7 year useful life) for financial reporting purposes. The cost of the asset is $100,000 and the salvage value used for depreciating purpose is $20,000. What is the difference in the current book value obtained using both the methods?
A. between $0 and $4,000
B. between $4,001 and $5,000
C. between $5,001 and $6,000
D. between $6,001 and $7,000
E. greater than $7,001

Problem 8

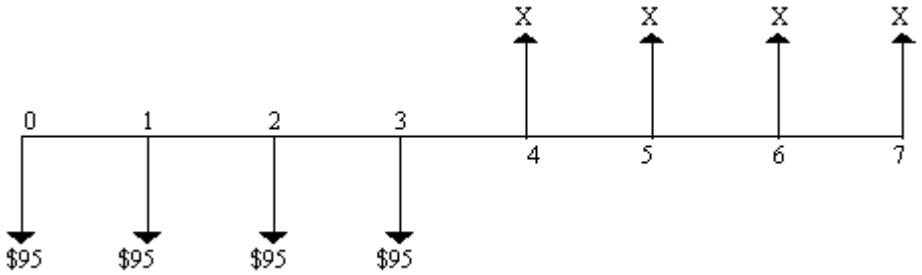
Which one of the following statements is correct?
A. Irrespective of the depreciation method adopted, total tax obligations will remain unchanged over the entire project life.
B. For a given cash flow, the present value always decreases as the interest rate increases.
C. Depreciation is a real cash expense since it represents the cost of doing business
D. When comparing mutually exclusive investments based on the rate of return principle, the incremental analysis need not be applied if all projects require the same initial investment.

Problem 9

You are planning to install a new CNC that costs $150,000. This machine has an estimated service life of 10 years and a net-after tax salvage value of $15,000. Its annual after tax operating and maintenance costs (considering depreciation tax shields) is estimated to be $35,000. To get a 18% rate of return on the investment (after-tax) , what will be the required minimum annual after-tax revenues?
A. between $0 and $60,000
B. between $60,001 and $65,000
C. between $65,001 and $70,000
D. between $70,001 and $75,000
E. greater than $75,001

Problem 10

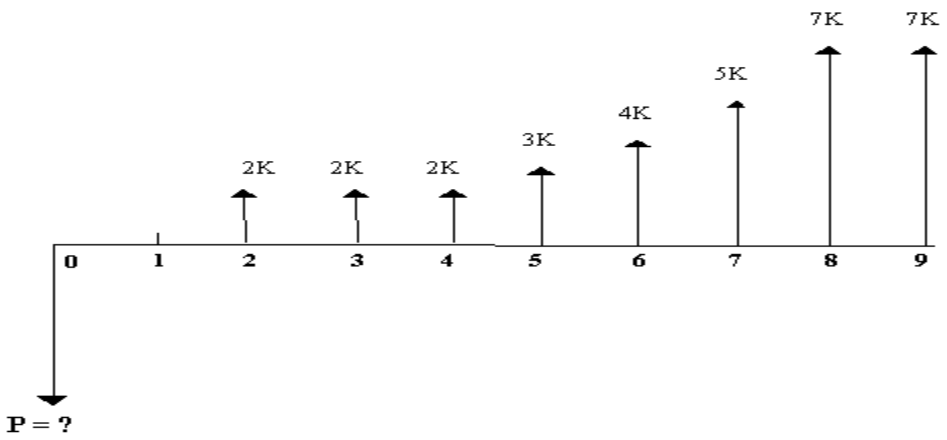
For the given cash flow below find the value of X that makes the inflow series equivalent to the outflow series. Use an interest rate of 8% compounded yearly.



A. X < $125
B. $125 <= X < $132
C. $132 < X <= $136
D. $136 < X <= $140
E. None of the above

Problem 11

If you wish to withdraw a series of cash amounts shown below, how much do you need to deposit today (n = 0) in an account that earns a 10% annual interest?



(Note: *K= $1,000*)
A. $2000(P/A, 10%, 8)(P/F,10%,1) + $1000(P/G,10%,6)(P/F,10%,3)
B. $2000(P/A,10%,8)(P/F,10%,1) + $1000(P/G,10%,3)(P/F.10%,4) + $5000(P/A,10%,2)(P/F,10%,7)
C. 2000(P/A,10%,8)(P/F,10%,1) + $1000(P/G,10%,4)(P/F,10%,3) +$5000(P/A,10%,2)(P/F,10%,8)
D. $2000(P/A,10%,8)(P/F,10%,1) + $1000(P/G,10%,5)(P/F,10%,4) + $1000(P/F,10%,8)
E. $2000(P/A,10%,8)(P/F,10%,1) + $1000(P/G,10%,4)(P/F,10%,3) + $1000(P/F,10%,8)

Problem 12

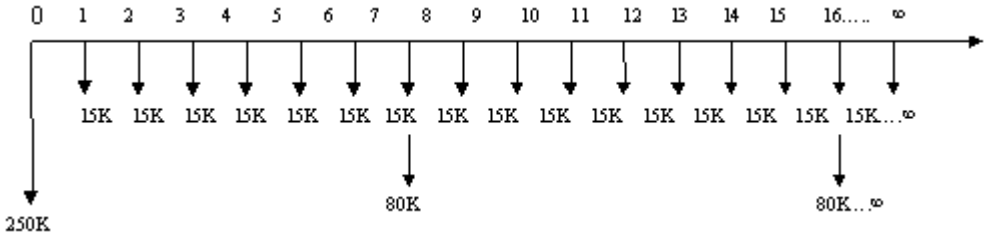
Micro-Tech Company is considering the expansion of its business by promoting a whole new product line. The expected annual taxable income before the expansion is estimated at $70,000. The expansion will bring in an additional annual revenue of $25,000, but it is expected to incur additional operating costs of $10,000 (excluding any depreciation expenses). This expansion requires the purchase of a new asset at a cost of $20,000. This new asset falls into the *MACRS* 3-year class. What is the expected annual net income for year 1 after the expansion?

(Note: *The Business expansion occurs at year 0 (now)*)
(Use the tax table given in Table 8.1, page 419)

A. less than $62,000
B. between $62,001 and $63,000
C. between $63,001 and $64,000
D. between $64,001 and $65,000
E. greater than $65,001

Problem 13

Company Z is considering the purchase of a new equipment for a replacement. Its initial cost is $250,000. The equipment requires an annual maintenance cost of $15,000. Also to be taken into account is an additional overhauling cost (at the end of every 7 years) of $80,000. The company plans to use the equipment for an infinite period. Find the capitalized equivalent cost of this investment at an interest rate of 12%.

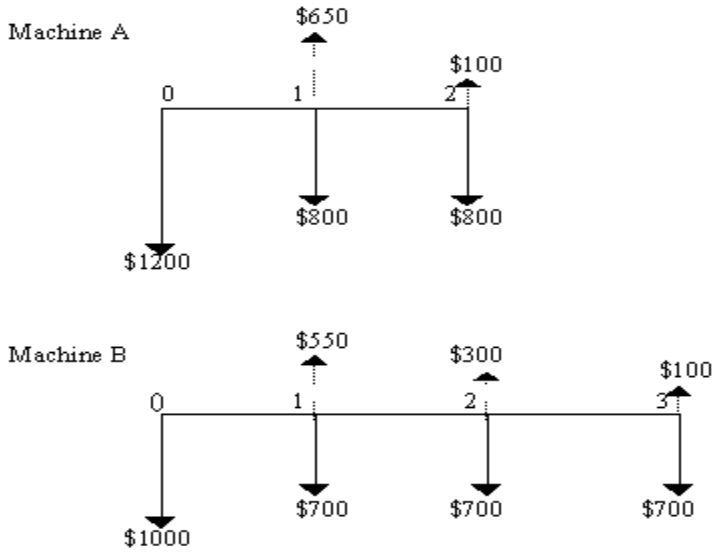


A. less than $439,000
B. between $439,001and $442,000
C. between $442,001 and $445,000
D. between $445,001 and $448,000
E. greater than $448,001

Problem 14

Consider two mutually exclusive machines, A and B. The service life of machine A is 2 years and that of machine B is 3 years. Assume that both machines will be available in the market for the same investment cost, salvage value and operating cost in the future. The expected salvage value at the end of each year is shown in dotted cash flows.

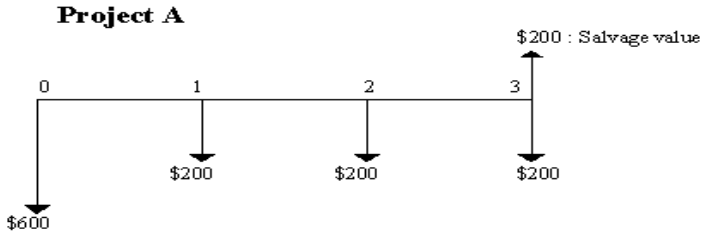
Given MARR=10%, which project should be selected if the study period is 3 years?

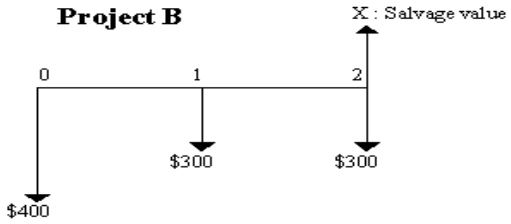


A. A
B. B
C. Either A or B
D. Neither A nor B
E. Information insufficient to analyze

Problem 15

Consider two mutually exclusive projects A and B.





Assume that each project can be repeated indefinitely with the same cash flows. Find the salvage value(X) of project B that would make both the projects equivalent at a MARR of 10%.
A. X < $200
B. $200 <= X < $250
C. $250 <= X < $300
D. $300 <= X < $350
E. X >= $350

Problem 16

For the project given below find the net cash flow at the end of year 4:

Project life (years)	5
Capital Investment at year 0	$100,000
Annual revenue	$75,000
Annual expenses (excluding depreciation)	$20,000
Salvage value at the end of the year	$10,000
Tax rate	40%
Depreciation method	7-year MACRS

A. less than $39,000
B. between $39,001 and $43,000
C. between $43,001 and $47,000
D. between $47,001 and $51,000
E. greater than $51,001

Problem 17

Suppose that in Problem No. 16, the firm borrows the entire capital investment at 10% interest rate over a period of 5 years. In addition , an amount of $20,000 is required as working capital at year 0 which will be recovered at the end of the project life. The required principal and interest payments for year 5 are as follows.

Principal Payment in year 5: $23,982
Interest Payment in year 5 : $2,398

What will be the net cash flow at the end of year 5?
A. less than $35,000
B. between $35,001 and $40,000
C. between $40,001 and $45,000
D. between $45,001 and $50,000
E. greater than $50,001

Problem 18

Assume that you deposited $100,000 in a savings account paying an interest of 6% compounded monthly. You wish to withdraw $2,000 at the end of each month. How many months will it take to deplete the balance?
A. less than 47 months
B. between 48 and 51 months
C. between 52 and 55 months
D. between 56 and 59 months
E. greater than 60 months

Problem 19

The city of Auburn needs to maintain its police cars. At present it has a contract with a local garage to do regular repair works and other routine maintenance works. The garage bills the city about $1,100 per car per year. As an alternative, the town is considering owning a small garage with two mechanics. The town has to spend $150,000 to buy the garage and furnish it. The annual cost of operating and maintaining the garage is about $60,000. In addition to the above expenses, another $500/car/year has to be spent on parts and other accessories. The garage has an estimated service life of 20 years and a salvage value of $50,000.

Find the minimum number of cars that the city has to maintain to justify the investment in the new garage. Use an interest rate of 6%.
A. less than 110
B. between 111 and 116
C. between 117 and 122
D. between 123 and 128
E. greater than 129

Problem 20

Consider the following two mutually exclusive investment alternatives

Alternative	**A**	**B**
Capital Investment	$100,000	$3,000
Annual Cost	$5,000	$17,500
Life	Infinite	Infinite
Salvage Value	$0	$0

At what MARR will you be indifferent between the two alternatives?
(Choose from the given choices the range within which the MARR falls).

A. 0%-8%
B. 8.1%-10%
C. 10.1%-12%
D. 12.1%-14%
E. 14.1%-25%

Solutions:

1. C
2. B
3. C
4. C
5. D
6. E
7. E
8. A
9. C
10. B
11. C
12. C
13. B
14. B
15. D
16. A
17. D
18. D
19. C
20. D

Set 1

Part I: True or False

Problem 1

When comparing mutually exclusive investments based on the rate of return principle, we don't need to apply the incremental analysis if all projects require the same initial investment:

A. True
B. False

Problem 2

The reason why the Congress allows the business to use a MACRS depreciation as opposed to the conventional ones is to reduce the business's tax burden over the project life.
A. True
B. False

Problem 3

Under an inflationary economy, a debt financing is always a more sensible option because you are paying back with cheaper dollars.
A. True
B. False

Problem 4

Depreciation is a real cash expense since it represents a cost of doing business:
A. True
B. False

Problem 5

In comparing mutually exclusive projects with unequal service lives, you always choose an analysis period equal to a required service period:
A. True
B. False

Problem 6

A project's future worth is equivalent to its terminal project balance at the specified interest rate:
A. True
B. False

Problem 7

Over the project life, a typical business will generate a greater amount of total project cash flows (undiscounted) by adopting a faster depreciation method:
A. True
B. False

Problem 8

No matter which depreciation method you adopt, your total tax obligations would remain unchanged over the project life:
A. True
B. False

Problem 9

For a given set of cash flow series, the present value of the cash flows will always decrease as you use a higher interest rate:
A. True
B. False

Problem 10

Under inflationary economy, a typical project would require less infusion of working capital over the project life:
A. True
B. False

Part II: Multiple Choice

In each question, select the nearest (closest) answer from the multiple choice list.

Problem 1

Eight years ago, a company purchased an injection molding machine at $40,000. It has been depreciated according to conventional straight-line method (book depreciation) over a 12-year service life. The estimated salvage value at the end of 12 years (from the purchase) was $4,000. This estimate is still good. The machine has a current market value of $15,000. If the machine is sold on the market, what would be the net proceeds (net salvage value after tax) from the sale? The firm's income tax rate is 40%. Any gains (or losses) will be taxed (or credited) at 40%.

A. $9,000
B. $11,000
C. $15,400
D. $13,800

Problem 2

You are considering purchasing a CNC machine which costs $150,000. This machine will have an estimated service life of 10 years with a net after-tax salvage value of $15,000. Its annual after-tax operating and maintenance costs (considering depreciation tax shields) are estimated to be $50,000. To expect a 18% rate of return on investment (after-tax), what would be the required minimum annual after-tax revenues?

A. $63,500
B. $82,740
C. $92,435
D. $94,568

Problem 3

A small machine shop has an estimated annual taxable income of $70,000 during 1997. Owing to an increase in business, the company is considering purchasing a new machine (in January of 1997) that will generate an additional (before tax) annual revenue of $100,000 over next 5 years. The new machine requires an investment of $120,000, which will be depreciated according to a 5-year MACRS property. What is the incremental income tax rate due to the purchase of the equipment in tax year 1997?

Use the corporate tax table (Table 8.1) on page 419.

A. 34%
B. 39%
C. 37.23%
D. 36.43%

Problem Statement for Questions 4 and 5

A father wants to save in advance for his 8-year-old daughter's college expenses. The daughter will enter the college 10 years from now. An annual amount of $20,000 in today's dollars (constant dollars) will be required to support the college for 4 years. Assume that these college payments will be made at the beginning of the school year. The future general inflation rate is estimated to be 5% per year, and the interest rate on the savings account will be 8% compounded quarterly (market interest rate) during this period.

Problem 4

What is the equal amount the father must save each quarter (in actual dollars) until his daughter goes to college?

A. $1,102
B. $2,000
C. $2,125
D. $2,063

Problem 5

If the father has decided to save only $1,000 each quarter, how much will the daughter have to borrow to support her college education at the beginning of sophomore year?

A. $4,120
B. $4,314
C. $4,000
D. $4,090

Problem 6

A corporation is considering purchasing a machine that will save $200,000 per year before taxes. The cost of operating the machine, including maintenance, is $80,000 per year. The machine will be needed for 6 years, after which it will have a zero salvage value. Alternative MACRS depreciation will be used assuming a 5-year class life. The allowed depreciation amount in percentage of the initial cost will be 10%, 20%, 20%, 20%, 20% and 10%, respectively. If the firm wants 15% rate of return after taxes, how much can they afford to pay for this machine? The firm's income tax rate is 40%.

A. $345,820
B. $363,640
C. $378,940
D. $300,569

Problem Statement for Questions 7 and 8

You are considering two options for manufacturing a typical product:

1. You continue to use an old machine now in use which was bought 8 years ago at $12,000. It has been fully depreciated but can be sold for $2,000. If kept, it could be used for 3 more years (remaining useful life), at the end of which time it would have no salvage value. The annual operating and maintenance costs amount to $10,000 for the old machine.

2. You purchase a brand new machine at an invoice price of $15,000 to replace the present equipment. Because of the nature of the product manufactured, it also has an expected economic life of 3 years, and will have a salvage value of $3,400 at the end of that time. With the new machine, the expected operating and maintenance costs amount to $3,000 for the first year and $4,000 in each of the next two years. The income tax rate is 35%. Any gains will also be taxed at 35%. The allowed depreciation amounts for the new machine are $2,143 during the first year, $3,673 during the second year, and $1,312 (with the half-year convention) during the third year.

Problem 7

What is the opportunity cost of not replacing the old machine now?

A. $700
B. $1,300
C. $2,000
D. $10,000

Problem 8

What is the incremental annual after-tax benefit of replacing the old machine at an interest rate of 15%?

A. $6,648
B. $7,069
C. $421
D. $960

Part III: Problem Solving

Problem 1

A proposed project which requires an investment of $10,000 (now) is expected to generate a series of five payments in constant dollars. It begins with $6,000 at the end of first year but increasing at the rate of 5% per year thereafter. Assume that the average inflation rate is 4% and the market interest rate is 11% during this inflationary period. What is the equivalent present worth of this investment?

Problem 2

Minolta Machine Shop purchased a computer-controlled vertical drill press for $100,000. The drill press is classified as a 7-year MACRS property. Minolta is planning to use the press for 5 years. Then Minolta will sell the press at the end of service life at $20,000. There is a working capital recovery of $22,000 at the end of 5 years and no further working capital is required in the future. The net annual revenues are estimated to be $110,000. If the estimated net cash flow at the end of year 5 is $72,000, what are the estimated operating and maintenance expenses in year 5. Minolta's income tax rate is 34%.

Problem 3

Harry Wilson, a mechanical engineer at Lehigh Manufacturing, has found that the anticipated profitability of a newly developed motion detector for its popular home security device product line can be estimated as follows:

$$\mathbf{NPW = 40.28V(2X - 11) - 77,860}$$

where V is the number of units produced and sold, and X is the sales price per unit. Harry also found that V parameter value could occur anywhere over the range of 1000 to 6000 units and the X parameter value anywhere between $20 to $40 per unit.

Suppose both V and X are statistically independent uniform random variables with the following means and variances:

E[V]= 3500
Var[V]= 2,083,333
E[X]= 30
Var[X]= 33

What is the mean and variance of the NPW?

If V and X are mutually independent discrete random variables with the following probabilities:

V		X	
Event	Probability	Event	Probability
1000	.30	20	.30
3000	.40	30	.50
6000	.30	40	.20

What is the probability that the NPW would exceed $7,000,000?

Solutions:
Part I